

Yuan Zhang

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PROFILE

Actuarial Studies graduate from UNSW with experience across insurance product design, pricing, claims and exposure analysis, and customer analytics. Strong quantitative foundation in actuarial modelling and data analysis, combined with a demonstrated ability to translate risk insights into practical product and commercial recommendations. Curious and collaborative, with a strong interest in credit risk, portfolio monitoring and data-driven decision-making in financial services.

EDUCATION & QUALIFICATIONS

University of New South Wales | Master of Actuarial Studies (Extension) | Sep 2024 – Sep 2026

- Actuaries of Tomorrow Scholarship, IFoA Foundation (2024)
- Relevant studies: actuarial statistics and insurance risk modelling, finance, business economics, risk management and actuarial data analytics

Xi'an Jiaotong-Liverpool University | Bachelor of Actuarial Science | Sep 2020 – Jun 2024

- University Academic Achievement Award (2022–23)
- IFoA exemptions: CM1, CM2 and CS1

INSURANCE & COMMERCIAL PROJECTS

Actuaries in Space: The Pricing Frontier | SOA Research Institute Student Research Case Study Challenge | 2026

- Analysed historical claims and prospective exposure data across four hazard lines to model expected losses, volatility and tail risk for a new commercial insurance opportunity
- Developed short- and long-term loss estimates and stress-tested portfolio performance and capital requirements under correlated risk scenarios, identifying key risk drivers and translating results into underwriting recommendations
- Designed coverage terms including deductibles, limits, triggers and exclusions to reflect portfolio risk profiles and accumulation risk.

Parametric Flood Insurance Product Innovation | Insurance Innovation Competition, UNSW | 2026

- Co-developed a parametric flood insurance product for Australian households, combining rainfall and river-level triggers with three-tier payouts to support faster recovery
- Analysed historical rainfall, river-gauge and flood-event data across NSW and Queensland to identify and calibrate regional payout triggers, and evaluated their reliability, basis risk and customer outcomes
- Designed a monitoring and recalibration framework to assess payout adequacy, false-positive and false-negative trigger performance, affordability and portfolio concentration, with scope for future multi-peril expansion

Freely Travel Insurance: Conversion Analysis | Product Analytics Project, UNSW | 2025

- Cleaned and validated two months of customer quote data, investigated data-quality issues and analysed quote-to-purchase conversion to identify key performance drivers and customer trends

- Co-developed an interactive dashboard and translated analytical findings into clear product and customer recommendations for non-technical stakeholders; the team report was selected as an outstanding cohort exemplar

PROFESSIONAL EXPERIENCE

Insurance Sales Director Assistant Intern | Xiamen Ping An Insurance | July 2022

- Supported policy issuance, sales tracking and client communications
- Compared insurance product structures and coverage against customer needs

Web Designer | Xiamen OUHUA | Jul – Aug 2024

- Built and maintained the company website, using visitor data to improve content, navigation and user experience

ADDITIONAL COMMERCIAL EXPERIENCE

Reverse Garbage: Participation & Market Analysis | Social Enterprise Consulting Project, UNSW | 2025

- Investigated barriers to participation using market research and stakeholder insights, identifying key factors limiting customer engagement and uptake
- Translated findings into practical recommendations to strengthen participation and customer value; the team project was awarded First Prize

SELECTED ANALYTICAL PROJECTS

Financial Time Series & Dependence Risk Analysis | Individual Actuarial Project | R | 2026

- Analysed approximately ten years of NVIDIA and AMD equity data using time-series and copula models to assess dependence and tail behaviour
- Estimated portfolio VaR and Expected Shortfall, validated model outputs against empirical risk measures, and identified downside concentration and tail-risk signals relevant to portfolio monitoring

Airbnb Predictive Modelling | Individual Machine Learning Project | R | 2025

- Built and evaluated predictive models through data cleaning, feature engineering, model comparison and out-of-sample validation to identify key drivers of the target outcome.
- Communicated model performance and practical implications in a concise analytical report selected as a high-quality exemplar for the cohort

TECHNICAL SKILLS

Analytics & Programming: R, Python, Excel, data visualisation, data cleaning and validation, statistical modelling, machine learning, data visualisation and dashboard development

Risk Analytic: Portfolio risk analysis, stress testing, VaR and Expected Shortfall, copulas, time-series analysis, claims and exposure analysis

Commercial: Risk reporting, insurance pricing, customer analytics, market research and stakeholder communication

Languages: Mandarin (native), English (professional working proficiency)